

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence
CODE: CSA17

COURSE

COURSE OUTCOME COVERED

***CO4:** Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

ANNEXURE A

ARTICLE REVIEW

Code of Conduct:

*I **NARAPPAGARI SRAVYA(192472192)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:SRAVYA

ANNEXURE A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a)** Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b)** Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a)** Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b)** Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors'

approach.

Task 3: Results & Critical Evaluation

- (a)** Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b)** Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c)** Discuss real-world applicability: Where can this technique be deployed?

What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

ANNEXURE A

ARTICLE REVIEW

Selected Research Article

Article Citation

Lynam, A. L., Dennis, J. M., Owen, K. R., Oram, R. A., Jones, A. G., Shields, B. M., & Ferrat, L. A. (2020).

“Logistic regression has similar performance to optimised machine learning algorithms in a clinical setting: application to the discrimination between type 1 and type 2 diabetes in young adults.”

Diagnostic and Prognostic Research, 4, Article 6.

DOI: 10.1186/s41512-020-00075-2

Publisher: Springer Nature

Year: 2020

PDF / DOI:

<https://doi.org/10.1186/s41512-020-00075-2>

Task 1: Article Summary

(a) Domain and ML Problem

The article belongs to the **healthcare and medical diagnosis** domain.

The research problem is to distinguish **Type 1 diabetes from Type 2 diabetes** in young adults who have already been diagnosed with diabetes.

The authors compared **Logistic Regression** with six machine learning algorithms:

- Gradient Boosting Machine
- MARS
- Neural Network

K-Nearest Neighbours

- Random Forest
- Support Vector Machine

The study used seven predictors: age, BMI, GADA antibodies, sex, total cholesterol, HDL cholesterol, and triglycerides.

(b) Key Objective and Research Gap

The main objective of the study was to determine whether advanced machine learning algorithms actually provide better clinical prediction than the traditional statistical method of Logistic Regression. The authors focused on distinguishing Type 1 from Type 2 diabetes because correct classification is important for selecting appropriate treatment.

The research gap was that previous studies comparing statistical models and machine learning models often used limited validation methods, default parameters, or did not perform external validation. They also commonly reported discrimination measures such as ROC-AUC without sufficiently evaluating calibration.

To address this gap, the authors compared seven models using the same predictors and applied hyperparameter tuning. They used nested cross-validation for internal validation and a separate external dataset for testing. They also evaluated calibration and clinical usefulness using decision curve analysis.

The study found that Logistic Regression performed similarly to more complex machine learning methods. In external validation, Logistic Regression achieved the highest ROC-AUC of 0.95. The findings suggest that a simpler statistical model can be highly effective when a small number of strong predictors are available.

Task 2: Methodology Analysis

(a) ML Techniques and Dataset

Techniques Used

The study evaluated seven classification algorithms:

1. Logistic Regression
2. Gradient Boosting Machine
3. MARS
4. Neural Network
5. K-Nearest Neighbours
6. Random Forest

7. Support Vector Machine

Logistic Regression was used as the traditional statistical learning model. The other algorithms represented different machine learning approaches.

Dataset

The training data came from the **Exeter cohort in the UK**.

- Initial cohort: 1,378 participants
- Final training dataset: **960 participants**
- Type 1 diabetes: **135 (14%)**
- External validation dataset: **504 participants**
- Type 1 diabetes in validation set: **105 (21%)**

Seven predictors were used:

- Age at diagnosis
- BMI
- GADA antibodies
- Sex
- Total cholesterol
- HDL cholesterol
- Triglycerides

Continuous variables were standardised, GADA was converted into positive/negative categories, and observations with missing predictor values were removed.

The models were tuned using grid search and 5-fold cross-validation. Nested cross-validation was used for internal performance estimation.

(b) Connection to CO 4

The article mainly relates to:

CO 4 – Statistical Learning Methods

Specifically, it demonstrates **Logistic Regression** for binary classification and compares it with other machine learning algorithms.

Methodological Strength

A major strength is the use of a **separate external validation dataset**. This provides a better test of whether the models generalise to new patients. The authors also used hyperparameter tuning and evaluated calibration and decision-curve performance.

Limitation

The study used only **seven predictors**, which limits the complexity of the information available to the machine learning models. The authors themselves identify this as a limitation. Also, all participants were of white European origin, which limits generalisation to other populations.

Task 3: Results & Critical Evaluation

(a) Key Results

The main evaluation metric was **ROC-AUC**.

Model	Internal Validation	External Validation
Logistic Regression	≥ 0.94	0.95
Other ML Models	≥ 0.93	≥ 0.92

All models performed well, with only small differences between them. Logistic Regression achieved the numerically highest ROC-AUC in external validation.

The study also found:

- Logistic Regression had satisfactory calibration.
- Neural Network and Gradient Boosting Machine had the best calibration.
- Logistic Regression and SVM showed good decision-curve performance.
- All models showed some decrease in performance during external validation.

(b) Critical Evaluation

The reported results are strong, but they should not automatically be considered universally applicable.

Potential Biases

1. The participants were all of **white European origin**.
2. Only seven predictors were used.
3. The external dataset came from another UK cohort, so performance in other countries is uncertain.
4. Missing observations were removed using complete-case analysis.

The decrease in ROC-AUC during external validation demonstrates why external validation is important.

Additional Experiments

The study could be strengthened by:

- Testing the models on more diverse populations.
- Using larger datasets.
- Including additional clinical variables.
- Performing subgroup fairness analysis.
- Conducting prospective clinical validation.
- Comparing calibration across different healthcare settings.

(c) Real-World Applicability

The approach could be deployed as a **clinical decision-support system** to help healthcare professionals distinguish Type 1 and Type 2 diabetes.

Possible applications include:

- Hospital decision-support systems
- Primary healthcare
- Electronic Health Records
- Diabetes screening tools
- Clinical risk calculators

Industry Challenges

Major challenges include:

- Availability of high-quality medical data.
- Patient privacy and security.
- Differences between hospitals and populations.
- Model bias.
- Regulatory approval.
- Integration with existing healthcare systems.
- Need for clinical validation.

Because Logistic Regression is relatively simple and interpretable, it may be easier to implement than complex black-box models.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

1. Logistic Regression Can Be Highly Effective

I learned that a traditional statistical model does not always perform worse than advanced machine learning.

CO 4 connection: Statistical Learning – Logistic Regression.

The study showed that Logistic Regression achieved an external ROC-AUC of **0.95**.

2. External Validation Is Important

A model may perform very well during internal validation but perform slightly worse on a new dataset.

CO 4 connection: Model evaluation and generalisation.

The study showed a reduction in performance during external validation, demonstrating the importance of testing models on independent data.

3. Interpretability Matters in Healthcare

The most complex model is not necessarily the best model for clinical deployment.

CO 4 connection: Statistical learning and model selection.

Logistic Regression is easier to interpret and understand than many complex machine learning models, making it attractive for healthcare applications.

(b) Proposed Extension

I would extend the research by testing the models on a **larger and more diverse international dataset** containing patients from different ethnic and geographical populations.

The same seven predictors could first be used, followed by additional clinical features such as blood pressure, glucose level, family history, and HbA1c.

The performance could then be compared using:

- ROC-AUC
- Accuracy
- Precision
- Recall
- F1-Score
- Calibration

This experiment is feasible because the existing methodology can be applied to a new dataset. It would improve understanding of whether the model generalises across different populations and could make the results more useful for real-world healthcare deployment.

Conclusion

This article demonstrates that **Logistic Regression can perform as well as more advanced machine learning algorithms** for a diabetes classification problem when a small number of strong predictors are available.

The most important lesson is that model complexity does not automatically mean better clinical performance. External validation, calibration, interpretability, and clinical usefulness are essential when selecting a machine learning model for healthcare.

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